CAMBRIDGE
International Education

Cambridge International AS & A Level

BIOLOGY **9700/52**

Paper 5 Planning, Analysis and Evaluation **February/March 2024**

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

© Cambridge University Press & Assessment 2024

[Turn over

PUBLISHED

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

© Cambridge University Press & Assessment 2024

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

© Cambridge University Press & Assessment 2024

6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations:

;	separates marking points
/	alternative answers for the same marking point
R	reject
A	accept
I	ignore
AVP	any valid point
AW	alternative wording (where responses vary more than normal)
ecf	error carried forward
underline	actual word underlined must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point
( )	the word / phrase in brackets is not required, but sets the context

© Cambridge University Press & Assessment 2024

Question	Answer	Marks
1(a)(i)	concentration of, sodium chloride (solution) **;**	**1**
1(a)(ii)	distilled water / deionised water / pure water (as diluent) **;** correct volumes of stock solution and water (in Table 1.1) **;**	**2**
1(b)(i)	*any **one** from:* 1 allows valid comparison (between onions in different sodium chloride concentrations) **;** 2 allows comparison as, initial / starting, mass (of onions) is, not constant / AW **;**	**1**
1(b)(ii)	*any **four** from:* 1 *idea that* the conclusion is only supported for onions left for hours **; ora** 2 (ref. to intercept of *x*-axis / 0% change in mass / water potential of onions when left for) 48 hours is 3.1% sodium chloride **;** 3 ref. to qualified experimental error **;** 4 ref. to anomalous result(s) on graph **;** 5 *idea that* line of best fit may give different value **;** 6 ref. to intermediate concentrations (of sodium chloride) not tested (between 1% and 5% sodium chloride) **;** 7 ref. to no, statistical analysis / standard error / 95% CI **;** 8 *idea that* not all onion cells have the same water potential **;**	**4**

© Cambridge University Press & Assessment 2024

Question	Answer	Marks
1(c)(i)	*any **six** from:* 1 use a minimum of five stated temperatures **;** 2 suitable method to, maintain a constant temperature (for each replicate / experiment) **;** 3 same / stated, variety / age / AW, of turnips **;** 4 same / stated, dimensions / AW, of turnip block **;** 5 ref. to apparatus for obtaining same size of turnip blocks **;** 6 for each temperature, place turnip blocks in, a beaker / AW, of (fresh / new) distilled water **;** 7 at each temperature, measure the initial mass **and** the final mass after a, set / stated, time **;** 8 for each temperature, use at least three different blocks **and** calculate a mean **;** 9 method to remove excess water from turnip block (before measuring final mass) **;**	**6**
1(c)(ii)	1 *horizontal axis label:* temperature, °C **and** *vertical axis label:* rate of osmosis **;** 2 appropriately shaped line **;**	**2**

© Cambridge University Press & Assessment 2024

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(c)(iii)</td><td>ref. to hazard and risk and precaution ;
<table>
<tr><td>hazard</td><td>Risk</td><td>Precaution</td></tr>
<tr><td>knife / blade / AW</td><td>injury</td><td>cut away from the hand / cut down on to a, board / tile / AW</td></tr>
<tr><td>turnip</td><td>allergy irritant</td><td>wear gloves / PPE</td></tr>
<tr><td>hot water / water-bath / method of heating</td><td>scald / burn / injury</td><td>wear, (heat resistant) gloves / PPE
use, tongs / AW</td></tr>
</table></td><td>1</td></tr>
</table>

Question	Answer	Marks
2(a)	to prevent the parent (fruit flies), reproducing / mating / AW, with the, first generation / offspring / AW **;**	**1**
2(b)	*to separate male and female fruit flies:* use hand lens / (binocular) microscope / AW **or** ref. to difference between male and female (fruit flies) **and** *to place fruit flies:* use a, paint brush / pooter / forceps / AW **;**	**1**
2(c)(i)	there is no difference between, the observed and expected, results **;**	**1**

© Cambridge University Press & Assessment 2024

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(c)(ii)</td><td>1 correct values of E ;

2 correct values of $(O–E)^2 / E$;

<table>
<tr><th>offspring phenotype</th><th>O</th><th>E</th><th>$(O–E)^2 / E$</th></tr>
<tr><td>dark red eyes</td><td>691</td><td>702</td><td>0.172
121 / 702</td></tr>
<tr><td>brown eyes</td><td>260</td><td>234</td><td>2.889
676 / 234 = 26 / 9</td></tr>
<tr><td>bright red eyes</td><td>225</td><td>234</td><td>0.346
121 / 234</td></tr>
<tr><td>white eyes</td><td>72</td><td>78</td><td>0.462
36 / 78 = 6 / 13</td></tr>
<tr><td>total</td><td>1248</td><td>(1248)</td><td>3.869</td></tr>
</table>

3 (χ^2 =) 3.869 ;</td><td>3</td></tr>
<tr><td>2(c)(iii)</td><td>1 ref. to correct critical value = 7.815 (p = 0.05 and df = 3) ;

2 calculated value of, χ^2 / 3.869, is less than 7.815 and null hypothesis is accepted ;</td><td>2</td></tr>
<tr><td>2(d)(i)</td><td>0.73 ;</td><td>1</td></tr>
</table>

© Cambridge University Press & Assessment 2024

Question	Answer	Marks
2(d)(ii)	*any **four** from:* *breeding experiment (**MAX 3**):* 1 ref. to the two genes (B / b and R / r), are **not** linked **;** 2 *idea that* dominant alleles code for (functional) carrier, proteins / polypeptides **;** 3 *idea that* in fruit flies with dark-red eyes entry of guanine and tryptophan (into organelle) allows pigments to be produced **;** 4 fruit flies with white eyes, (have the genotype) bbrr / homozygous recessive (for both genes) **or** fruit flies with dark-red eyes are, B_R **;** *chromatogram (**MAX 3**):* 5 fruit flies with dark-red eyes, have 4 pigments (in their eyes) **;** 6 fruit flies with white eyes, have no pigments (in their eyes) **;** 7 pigment 4 is drosopterin **;** 8 *idea that*, ommochrome / brown pigment, is insoluble (in the chromatography solvent) **or** *idea that* none of the pigments in the chomatogram are, ommochrome / brown pigment **;**	**4**

© Cambridge University Press & Assessment 2024